

NGUYEN VAN TRUONG

AI PROGRAMMING INTERN

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PROFESSIONAL SUMMARY

Final-year Computer Science student targeting an AI Programming internship, with hands-on practice in Python, PyTorch, FastAPI/RESTful API, SQL basics, and data processing. Built an end-to-end PPE monitoring prototype and an academic RAG prototype for financial-report question answering. Interested in working with mentors and software teams on practical AI projects.

TECHNICAL SKILLS

Programming: Python (primary); C/C++ and Java (academic foundation)
AI/ML & Computer Vision: PyTorch, Torchvision, Ultralytics YOLO, OpenCV, MobileNetV3, ByteTrack
Retrieval & NLP: FAISS, Sentence Transformers, text preprocessing, chunking, embedding-based retrieval
Data & Backend: SQL basics, NumPy, Pandas, PySpark; FastAPI, RESTful API, Pydantic
Workflow: Git/GitHub, Google Colab; AI-assisted research/debugging; dataset provenance, grouped splits, artifact hashing

SELECTED PROJECTS

Computer Vision System for PPE Compliance Monitoring 2026

Python, PyTorch, YOLO26s, MobileNetV3, ByteTrack, OpenCV, FastAPI, RESTful API

- Designed a modular fixed-camera pipeline for a 0-15 m construction-site zone, combining person detection, tracking, head/torso crops, PPE classifiers, and temporal status stabilization.
- Integrated YOLO26s, ByteTrack, and MobileNetV3 headwear/vest classifiers with fail-closed handling for missing or uncertain safety equipment.
- Integrated model inference with a FastAPI/RESTful API service and local UI for image uploads, annotated video playback, and live frame requests through a Colab GPU backend.
- Added source provenance, perceptual-hash deduplication, grouped splits, class contracts, artifact hashes, and smoke-test evidence on 10 images with 51 detections.

RAG Chatbot for EDGAR Financial Reports (Academic Prototype)

Python, FAISS, Sentence Transformers, LLaMA, document chunking

- Built a retrieval-augmented question-answering prototype over EDGAR 10-K/10-Q financial reports, covering preprocessing, chunking, embeddings, vector indexing, retrieval, and LLM answer generation.
- Experimented with single-hop and multi-hop retrieval strategies and reviewed answer quality with automatic metrics and LLM-based evaluation during coursework.

Traffic Sign Detection from Video

Python, OpenCV, NumPy

- Built a video pipeline using HSV segmentation, morphological operations, contour filtering, and normalized cross-correlation template matching.
- Rendered bounding boxes and labels with OpenCV and evaluated multiple sign types across different lighting and background conditions.

Transaction Analysis and Recommendation System

Python, SQL basics, PySpark, Pandas

- Processed transactional data with PySpark RDD/DataFrame APIs and SQL-style filtering, grouping, and aggregation; implemented Apriori, PCY, and FP-Growth for frequent-itemset mining.
- Evaluated ALS/SVD recommendation methods in a reproducible preprocessing and evaluation workflow.

EDUCATION

TON DUC THANG UNIVERSITY | B.Sc. in Computer Science | 2021-2026 (Expected)

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Big Data Processing, Data Structures and Algorithms

LEADERSHIP

Secretary, Youth Union - Class 21H50302 | 2021-Present

- Coordinated class activities, supported internal events, and contributed to website and communication content.